QUESTIONS

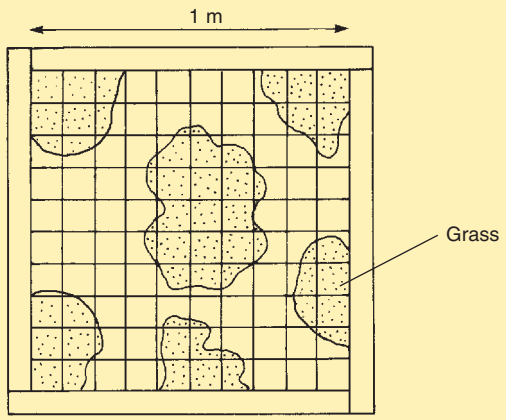
1. Explain each of these terms:
   1 Ecosystem
   2 Habitat
   3 Profile map
   4 Direct search
   5 Beating tray
   6 Pooter
   7 Tullgren funnel
   8 Mammal trap
   9 Plankton net
   10 Key
   11 Quadrat
   12 Capture-recapture method
   13 Random sample
   14 Grid
   15 Environmental gradient
   16 pH
   17 Light intensity
   18 Humidity
   19 Humus
   20 Salinity
   21 Exposure
   22 Cup anemometer
   23 Hygrometer
   24 Results

2. Distinguish between the following terms:
   (a) Biotic and abiotic factors
   (b) Qualitative and quantitative study
   (c) Identify and estimate
   (d) Fauna and flora
   (e) Pitfall trap and cryptozoic trap
   (f) Natural and artificial classification
   (g) Grid and gradient
   (h) Adaptations and interdependence
   (i) Percentage cover and frequency
   (j) Results and conclusions
   (k) Assess and analyse

3. (a) Draw a simple map of an ecosystem you have studied.
   (b) Describe the methods you used in obtaining information on the distribution of the plants and animals in the ecosystem. Name five plants and five animals in the ecosystem.
   (c) Give an example of a food web. Your example should contain at least eight components.
   (d) Name the producers and the primary consumers from your web.
   (e) At which level in the diagram is the greatest amount of energy to be found?
   (f) Give one example of competition from among the plants or animals listed.
   (g) Give two ways in which bacteria are of benefit in the ecosystem.
   (h) In the case of two of the plants and two of the animals listed point out some special features which would account for their success in the ecosystem.
   (i) State briefly what might happen if the carnivores were removed from the ecosystem.

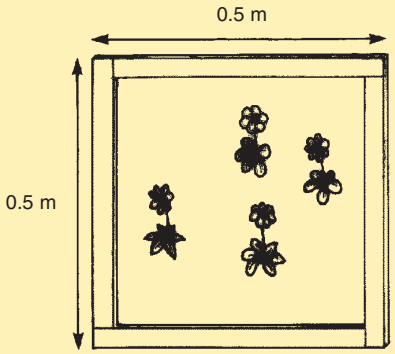
4. Describe each of the following (in words and by labelled diagrams) and write a note on their use in any ecosystem: (a) pooter, (b) belt transect, (c) quadrat, (d) Tullgren funnel, (e) beating tray, (f) pitfall trap.

5. Calculate the percentage cover of the grass in this diagram.



6. Your friend tells you that there are more minnows (small fish) in his pond than in yours. What could you do to find out if he is right?

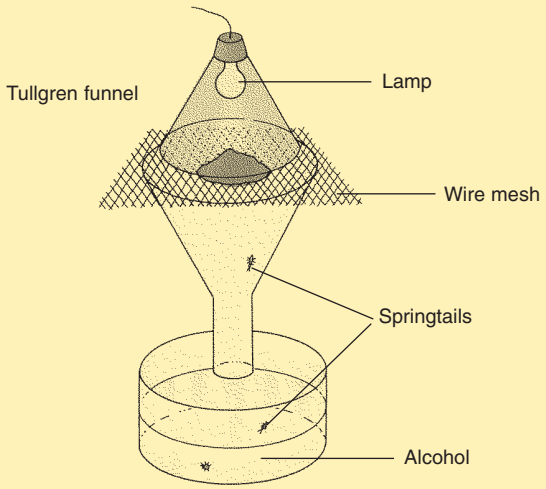
7. In a field 100 m by 100 m, there were too many daisy plants to count, so instead a student sampled the field using a quadrat (0.5 m square) to get an estimation of the number. The quadrat was thrown ten times, at random and the number of plants recorded as shown.



QUADRAT NUMBER	1	2	3	4	5	6	7	8	9	10
Number of daisy plants	4	0	6	11	3	2	10	0	0	1

(a) Draw a bar chart to illustrate these results.
(b) Calculate:
(i) the total number of daisy plants,
(ii) the average number of plants per quadrat,
(iii) the area of the quadrat,
(iv) the area of the whole field,
(v) an estimate of the total number of daisy plants in the field.

8. Robert noticed some tiny white insects called springtails in the soil below an oak tree. Springtails feed on fungi which decay the leaves. He investigated the distribution of the insects around the tree using five quadrats to the north and five to the south of the tree. He took an equal sample of soil from each quadrat. He counted the number of springtails in each sample using the apparatus below.



(a) (i) Why did Robert take an equal sample of soil from each quadrat?
(ii) Describe how he used the apparatus shown.
(b) The results of the investigation are shown in the table below.

	NORTH OF TREE					SOUTH OF TREE				
Number of springtails per soil sample	20	30	45	10	25	10	15	20	5	5

Calculate the average number of springtails per sample in the (i) south and (ii) north.
(c) Suggest possible reasons for the difference.

9. The number of animals in a habitat were studied using the 'capture-recapture' technique. The results are shown in the table.

ANIMALS	NO. IN 1ST CATCH (C_1)	NO. IN 2ND CATCH (C_2)	NO. OF MARKED ANIMALS IN 2ND CATCH (M_2)
Millipedes	40	50	25
Centipedes	50	60	20
Beetles	16	15	6
Woodlice	100	80	40
Snails	20	15	10

(a) Draw a bar chart for each animal population.
(b) Calculate the population of each animal in the habitat.
(c) Why should the animals not be marked with bright, permanent paint?

10. A lawn 50 m^2 was surveyed by a student using a 1 m^2 quadrat to obtain an estimate of the number of earthworms in the lawn. He threw a pencil over his shoulder and placed the quadrat with the pencil at the centre. The results of his findings are set out in the table below.

QUADRAT NUMBER	1	2	3	4	5
Number of earthworms	25	18	27	20	10

(a) Why did he throw the pencil over his shoulder?
(b) Why did he use the quadrat five times?
(c) It is known that earthworms mainly feed on dead leaves. Which one of the quadrat throws would most likely contain most dead leaves?
(d) Calculate the average number of worms per quadrat.
(e) Calculate the total number of worms in the lawn as estimated by this method?

(f) Explain how you would cause the earthworms to come to the top of the soil.

(g) Suggest one reason why earthworms are of importance in an ecosystem.

11. (a) Climatic factors can affect the distribution of organisms in an ecosystem.
Give three examples of these factors and how they might influence the distribution of an organism. Give one example to illustrate how a knowledge of such factors could be of use to a farmer or fisherman.

(b) Edaphic (soil) factors can affect the distribution of organisms in an ecosystem.
Give three examples of these factors and how they might influence the distribution of an organism.

(c) Topographic factors can affect the distribution of organisms in an ecosystem.
Give three examples of these factors and how they might influence the distribution of an organism.

12. (a) (i) Distinguish clearly between the terms food chain and food web.
(ii) 'The number of steps or links in a food chain sequence is limited, usually to four or five.' Comment on this statement in relation to the efficiency of energy transfer in an ecosystem.

(b) Here is a key to common invertebrates found in gardens.

1	Animals without legs	2
	Animals with legs	5
2	Animals divided into rings or segments	3
	Animals not divided into rings or segments	4
3	Tapered body	Fly larvae
	Cylindrical body	Worms
4	Coiled shell	Snails
	No shell	Slugs
5	Three pairs of legs	Insects
	More than three pairs of legs	6
6	Four pairs of legs	7
	More than four pairs of legs	9
7	Body divided into two distinct parts	Spiders
	Body not clearly divided into two distinct parts	8
8	Legs much longer than body	Harvestmen
	Legs much shorter than body	Mites
9	Seven pairs of legs	Woodlice
	More than seven pairs of legs	10
10	All legs alike	11
	Front three pairs of legs jointed, hind legs stumpy	Caterpillars
11	One pair of legs per segment	Centipedes
	Two pairs of legs per segment	Millipedes

(i) Use the above key to identify the following animal: its body is divided into twelve segments; the front three segments have segmented legs, and there are five pairs of stumpy legs on other segments. Show how you arrive at the answer.

(ii) Name three animals mentioned in the key which have eight legs.

(iii) What information does the key give you about snails?

13. A farmer wanted to find out the density of clover in one of his fields. He threw a 1 m^2 quadrat at random twenty times and obtained the results shown in the table below.

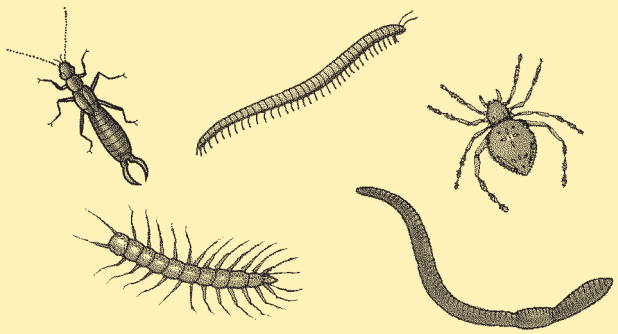
Throw of quadrat	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
No. of clover plants	5	0	3	1	12	8	8	8	22	13	6	3	7	10	12	4	10	13	12	15

(a) Calculate the total number of clover plants in the twenty throws of the quadrat.

(b) Calculate the average number of clover plants per square metre.

(c) What evidence is there in the table above to suggest that the clover plants are not evenly distributed in the field?

14. A sample of soil and decaying leaves was collected from a local wood.

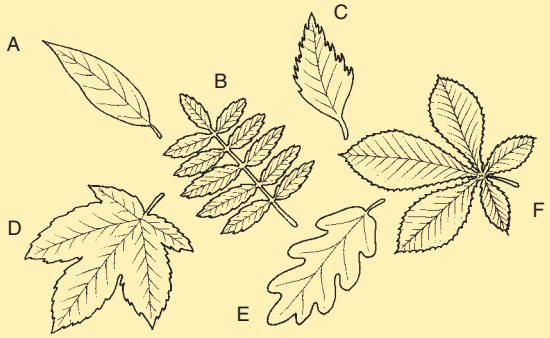


(a) The diagram shows some animals found in this sample. Use the key below to identify the animals.

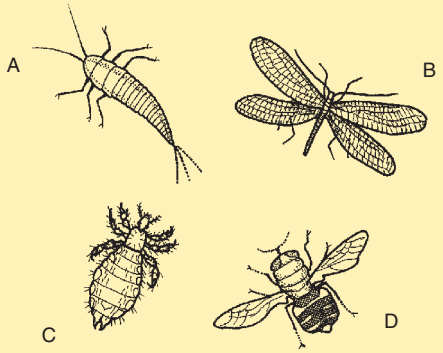
1.	8 or more walking legs present	2
	Fewer than 8 walking legs present	4
2.	Legs shorter than the body	3
	Legs longer than the body	*Lycosa*
3.	All body segments have one pair of legs	*Geophilus*
	Some body segments without legs	*Scutigerella*
4.	'Pincers' on tip of abdomen	*Forficula*
	No 'pincers' on tip of abdomen	*Enchytraeid*

15. Use the key below to identify the leaves of these common trees:

1	Leaves simple (not divided into leaflets)	Go to 2
	Leaves compound (divided into leaflets)	Go to 5
2	Leaves divided into 5 lobes	Sycamore
	Leaves not divided into 5 lobes	Go to 3
3	Edge of leaf smooth	Privet
	Edge of leaf not smooth	Go to 4
4	Edge of leaf toothed	Silver birch
	Edge of leaf with rounded lobes	Oak
5	Leaflets arranged like fingers of a hand	Horse chestnut
	Leaflets in pairs on leaf axis	Mountain ash



16. (a) Identify the animals A, B, C and D using the key below. The key can be used for six animals but you are to identify only the four that are shown. Write the letter of each animal in the box next to its name.



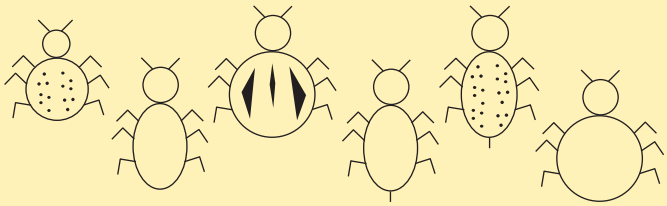
1.	Wings present	2	
	Wings absent	4	
2.	Antennae as long as body	*Chrysopa septempunctata*	☐
	Antennae shorter than body	3	
3.	One pair of wings	*Stratiomyia potamida*	☐
	Two pairs of wings	*Aphis rumicis*	☐
4.	Body more than three times longer than wide	*Lepisma saccharina*	☐
	Body not more than three times longer than wide	5	
5.	Legs longer than body	*Tegenaria domestica*	☐
	Legs shorter than body	*Pediculus humanus*	☐

(b) Why is size rarely used in keys?

(c) (i) Animal C is a parasite. What is meant by the term 'parasite'?
(ii) This parasite lives on the skin of a mammal. Describe one feature shown in the diagram of C which could help it to live as a parasite.

17. A pitfall trap was used to catch 12 ground beetles, which were marked and released. After two days a second catch gave 16 beetles. Of these 4 were marked. Calculate the total population of beetles in the habitat.

18. When Karen visited the island of Santorini, she discovered some insects. She made some drawings and brought them back home. Here are the sketches:



(a) Make up a name for each insect.

(b) Construct a key so that other visitors to the island may be able to identify them.

19. (a) Draw a diagram to illustrate a pyramid of numbers in an ecosystem. Place the words carnivore, herbivore and producer in their correct positions on your diagram. For each of the trophic levels on the pyramid, name an organism from the ecosystem you have studied.

(b) Describe, with the aid of diagrams, any two of the following and explain the purpose of each of the two you have chosen in a practical study of an ecosystem: (i) quadrat, (ii) pitfall trap, (iii) plankton net, (iv) pooter, (v) beating tray.

20. (a) Name an ecosystem you have studied and draw a simple outline map of that ecosystem. Indicate on the map at least two physical features. Give an example, from the ecosystem, of: (i) competition between animals, (ii) competition between plants.

(b) While on a field trip, your teacher asks you to take a small sample of soil back to the laboratory to see if there are any small animals in it. How would you carry out this experiment in the laboratory?

21. The numbers of plants and animals were counted at 3 different places (A, B, C) in a pond. The results are shown in the table.

PLACE / NUMBERS COUNTED	Small fish	Water flea	Duck	Small plants (algae)
A	4	500	1	10,000
B	1	700	0	8,000
C	3	680	1	7,000
Totals				

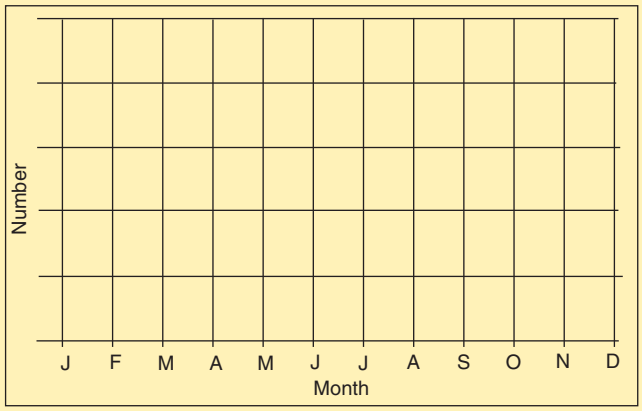
(a) Copy the table and complete the totals for each organism.
(b) Use the names in the table to draw a pyramid of numbers.
(c) Explain why all the plants in the pond were found near the surface.
(d) Plants are food for pond animals. Give two other ways in which these animals depend upon plants.

22. Soil is composed of several constituents including water and humus.
(a) In an experiment to determine the amount of water in soil, a sample of fresh soil from the garden was weighed and was found to be 90 g. It was then dried until all the water was removed, at which point it weighed 75 g. Calculate the weight of water and the percentage of water in the soil sample.
(b) The dry soil was then burned to remove all the humus, at which point it weighed 45 g. Calculate the weight of humus and the percentage of humus in the original soil sample.

23. (a) Name an ecosystem that you have studied. Name three plants that are normally present in this ecosystem.
(b) Answer the following in relation to one of the plants you have named in (a).
Describe how you carried out a survey to find out how many plants of this species were present in your study area.
(c) How did you present the results of the survey?
(d) Suggest one possible error that may affect the results of the survey.

(LC OL Sample Paper)

24. (a) Name an ecosystem that you have studied. Name three animals that are normally present in this ecosystem.
(b) Select one of the animals that you have named in (a) and answer the following questions in relation to it.
(i) Which animal have you selected?
(ii) State two features that allowed you to identify the animal.
(iii) Name an organism on which this animal normally feeds.
(iv) Explain how you attempted to find out how many of these animals were present in the ecosystem.
(v) Using the axes below, draw a graph to show how you would expect the numbers of this animal to vary in the ecosystem in the course of a year.



(LC OL Exam Paper)

HIGHER LEVEL

25. (a) A 2.5 hectare field was surveyed for clover plants using the quadrat method. A quadrat of side 0.5 m was used. The results of the survey are shown in the table below.

Quadrat Number	1	2	3	4	5	6	7	8	9	10
Number of Plants	6	2	2	2	4	0	2	5	5	2

Estimate the number of clover plants in the field. (1 hectare = 10,000 m^2)
(b) A survey of field mice was carried out in the field mentioned in (a) using the capture-recapture method. The field mice were caught using small mammal traps which were set at random points in the field. Forty field mice were captured, tagged and released at their capture points. One month later the traps were again set at the same locations and forty field mice were caught. Five of these were found to be tagged. Estimate the population density of the field mice in numbers per hectare.

(LC HL Sample Paper)